Paleomagnetism of the Maringouin and Shepody formations, New Brunswick: a Namurian magnetic stratigraphy[1]

V. J. DiVenere and N. D. Opdyke
Department of Geology, University of Florida, Gainesville, FL 32611, U.S.A.
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Two hundred thirty-five oriented cores were taken in a stratigraphic sequence of 575 m of red sediments from the Namurian Maringouin and Shepody formations. These samples were taken from the south-dipping limb of the Maringouin Anticline on the Maringouin Peninsula in eastern New Brunswick. An additional 59 samples of the Maringouin and Shepody formations were taken from the north-dipping limb for separate fold tests. Progressive thermal demagnetization to temperatures as high as 700°C permitted the resolution of an intermediate-temperature, syn-folding B component and a high-temperature, pre-folding C component. The C-component mean for the Maringouin Formation yields a north paleopole at 32.4°N, 121.3°E. The Shepody Formation mean yields a paleopole at 35.7°N, 124.3°E. The circles of confidence (α_{95}) for the Maringouin and Shepody formations are 3.8° and 4.6°, respectively. Plotting sample polarity versus stratigraphic position yields the first coherent magnetic stratigraphy from the late Paleozoic of North America, with five discrete reversed-polarity and five discrete normal-polarity zones. The magnetic field in Namurian times had approximately 50% normal and 50% reversed polarity.

Dans une séquence stratigraphique de 575 m de sédiments rouges des formations de Maringouin et de Shepody, d'âge namurien, 235 carottes orientées ont été prélevées. Ces échantillons proviennent de forages dans le flanc incliné vers le sud de l'anticlinal de Maringouin, localisé sur la péninsule de Maringouin dans le Nouveau-Brunswick oriental. Un groupe supplémentaire de 59 échantillons a été prélevé dans les formations de Maringouin et de Shepody sur le flanc incliné vers le nord pour y appliquer les tests du pli. La désaimantation progressive par chauffage à des températures aussi élevées que 700°C a permis de résoudre une température intermédiaire, contemporaine de la déformation et correspondant à la composante B, et une température élevée, antérieure à la déformation et représentant la composante C. La moyenne de la composante C pour la Formation de Maringouin fournit un paléopôle nord à 32,4°N, 121,3°E. La moyenne de la Formation de Shepody fournit un paléopôle à 35,7°N, 124,3°E. Les cercles de confiance (α_{95}) pour les formations de Maringouin et de Shepody sont 3,8° et 4,6°, respectivement. Le graphique de la polarité des échantillons contre la position stratigraphique fournit la première stratigraphie magnétique cohérente du Paléozoïque supérieur de l'Amérique du Nord, incluant cinq zones distinctes de polarité inverse et cinq zones distinctes de polarité normale. Le champ magnétique durant les temps du Namurien comprenait approximativement 50% de polarité normale et 50% de polarité inverse.

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Introduction

Paleomagnetic studies in the Maritimes basins were carried out by J. Roy and his colleagues and were discussed in a series of papers, including those by Roy and Robertson (1968) and Roy and Park (1969, 1974) on the Carboniferous Maringouin and Shepody formations of New Brunswick. These studies showed that sites containing both normal and reversed magnetization were present in these formations. The dual-polarity magnetization was shown to have been acquired before folding, which was considered Pennsylvanian.

The present study of the Maringouin and Shepody formations was undertaken in an attempt to uncover the reversal pattern of the Earth's magnetic field prior to the onset of the Permo-Carboniferous reversed superchron (PCRS). It was previously pointed out by Roy (1969) and Roy and Morris (1983) that these sediments must have been deposited before the PCRS, or the Kiaman interval of Irving and Parry (1963).

Geologic setting

Carboniferous sediments in the Maritimes were deposited in isolated sedimentary basins that formed during the Appalachian Orogeny. The Maringouin Formation and the overlying Shepody Formation were deposited in the Carboniferous Cumberland structural basin. A well-exposed, continuous sequence of these strata is found as coastal cliffs on the eastern shore of the Maringouin Peninsula of New Brunswick. The strata form the Maringouin Anticline, which trends more or less east–west across the Maringouin Peninsula. The peninsula separates Shepody Bay from Cumberland Basin in eastern New Brunswick (Fig. 1).

Maringouin Formation

The Maringouin Formation is composed of red, calcareous mudstone and siltstone, with some interbeds of fine-grained sandstone (Belt 1965). According to Gussow (1953), the formation represents a "transition from the marine Windsor beds to the overlying red continental sandstone of the Shepody Formation." Weeks (1963) reported the age of Maringouin Formation as Late Mississippian. The formation is thought to be at least as old as microspore zone "a" of M. S. Barss (Belt 1965), and Belt gave its age as Namurian A.

Shepody Formation

The overlying Shepody Formation is composed of red interchannel lutites and thin-bedded sandstones; there are also numerous channel units of greenish gray to red, fine to medium sandstones and limestone-pebble conglomerates (Belt 1965). Microspores are found in the formation, and Bell (1944) cited a very early Namurian assemblage of plant fossils. Fossil micro-

[1]This paper is dedicated to Jean Roy, whose pioneering studies in the Carboniferous and Permian in the Maritimes region were of great importance to North American paleomagnetism.

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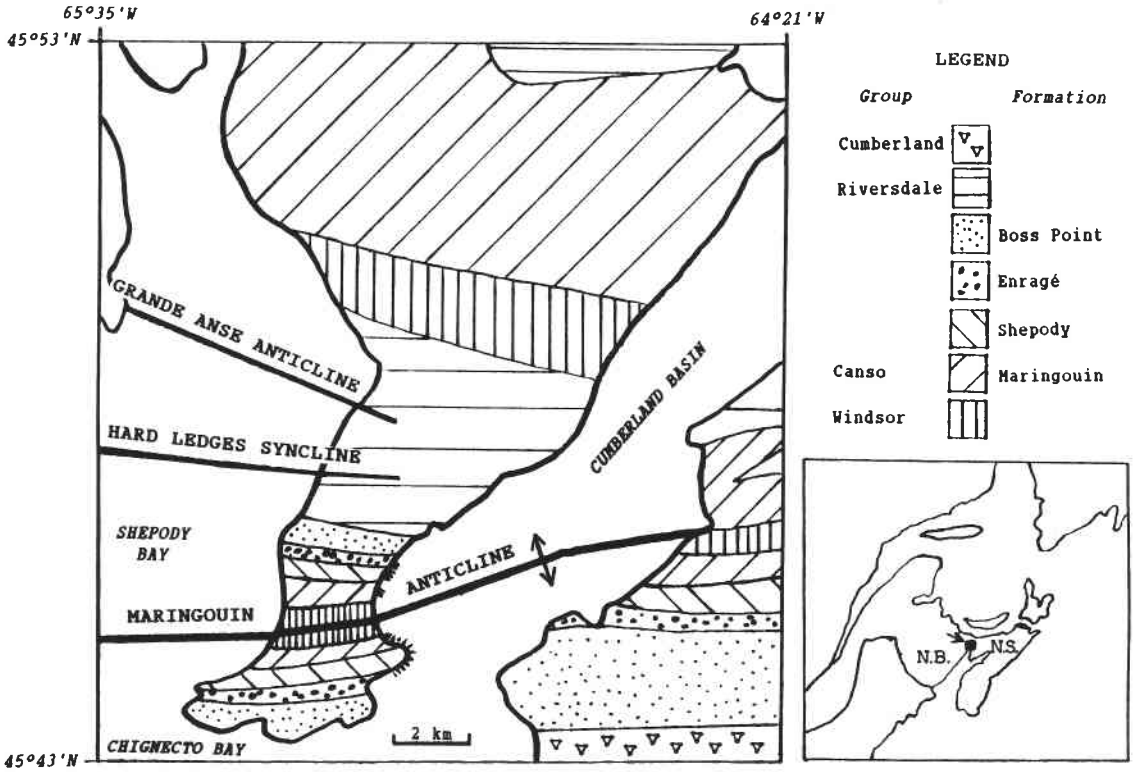


FIG. 1. Maringouin Peninsula geology. Modified from Roy (1977). Hatched areas on southeast coast of the peninsula show sampled section.

spores of microspore zone "c" (Namurian C) were identified by Barss (Belt 1965) in a laterally equivalent portion of the upper Shepody Formation in Nova Scotia, 5 mi (1 mi = 1.6 km) to the east of the Maringouin Peninsula.

Procedures

Sampling

A portable, gasoline-powered drill was used to obtain 2.25 cm diameter samples with lengths to 12 cm. Samples were later cut to 1.25–1.5 cm lengths. Sample orientation was performed with a Brunton compass mounted on a core-orienting device. Strike and dip measurements of bedding orientations throughout the sequence were taken about every 5 samples with a Brunton compass. The section was measured during sample collection, and each sample was located in the lithostratigraphy.

In 1986–1988, 235 samples were taken in a stratigraphic sequence of 575 m of the Maringouin and Shepody formations from the south-dipping limb of the Maringouin Anticline on the eastern shore of the Maringouin Peninsula. Single samples were taken in sequential order at an average sampling interval of 2.4 m. Sampling density was greater in the fine sediments of the Maringouin Formation than in the Shepody Formation because of the occurrence in the latter of channel-sandstone interbeds, from which no samples were obtained. The outcrop on the south limb consists of a continuous section comprising the upper 285 m of the Maringouin Formation and 290 m of the Shepody Formation. The base of the Maringouin Formation was not seen; the base of the section is placed at its first outcrop along the beach. This study places the Maringouin–Shepody boundary at the first promontory-forming green sandstone. The Shepody Formation, from the first massive sandstone to its upper contact with the Enragé Formation, is complete.

Twenty-six samples of the Maringouin Formation and 33 samples of the upper Shepody Formation were taken from the north-dipping limb of the anticline on the eastern shore of the peninsula for fold tests. The Maringouin Formation consists of a discontinuous series of outcropping beds without an upper or lower contact for reference. The sampled outcrop of the Shepody Formation on the north limb begins approximately 150 m from the top of the formation as determined by its contact with the Enragé Formation. The Shepody beds are more continuous than the Maringouin beds on the north limb because of the coarser nature of the sediments.

Remanent magnetization

The remanent magnetization of the samples, beginning with the natural remanent magnetization (NRM), was measured in an SCT two-axis cryogenic magnetometer at the University of Florida. Storage and all procedures were performed inside a double-walled steel room (Scott and Frohlich 1985) with an average field of 50–250 nT; the thermal demagnetizer and magnetometer were housed inside nested mu-metal cylinders, and samples were placed temporarily inside a mu-metal cylinder after thermal treatment and before they were measured.

Generally, samples were treated in 100°C steps to 600°C; intermediate steps in the 500°–600°C range were used on some samples. Samples were subjected to 10°C steps between 600° and 660°C and to 5°C steps thereafter. The Maringouin Formation and the lowest Shepody Formation collected from the south limb in 1986 had only 30 pilot samples subjected to this full treatment; for the other 124 samples we omitted the 100°–500°C range and skipped from NRM directly to 550°C. For the 81 Shepody Formation samples taken from the south limb in 1987 we omitted the 100°C step.

Principal components

For each sample, principal components were calculated

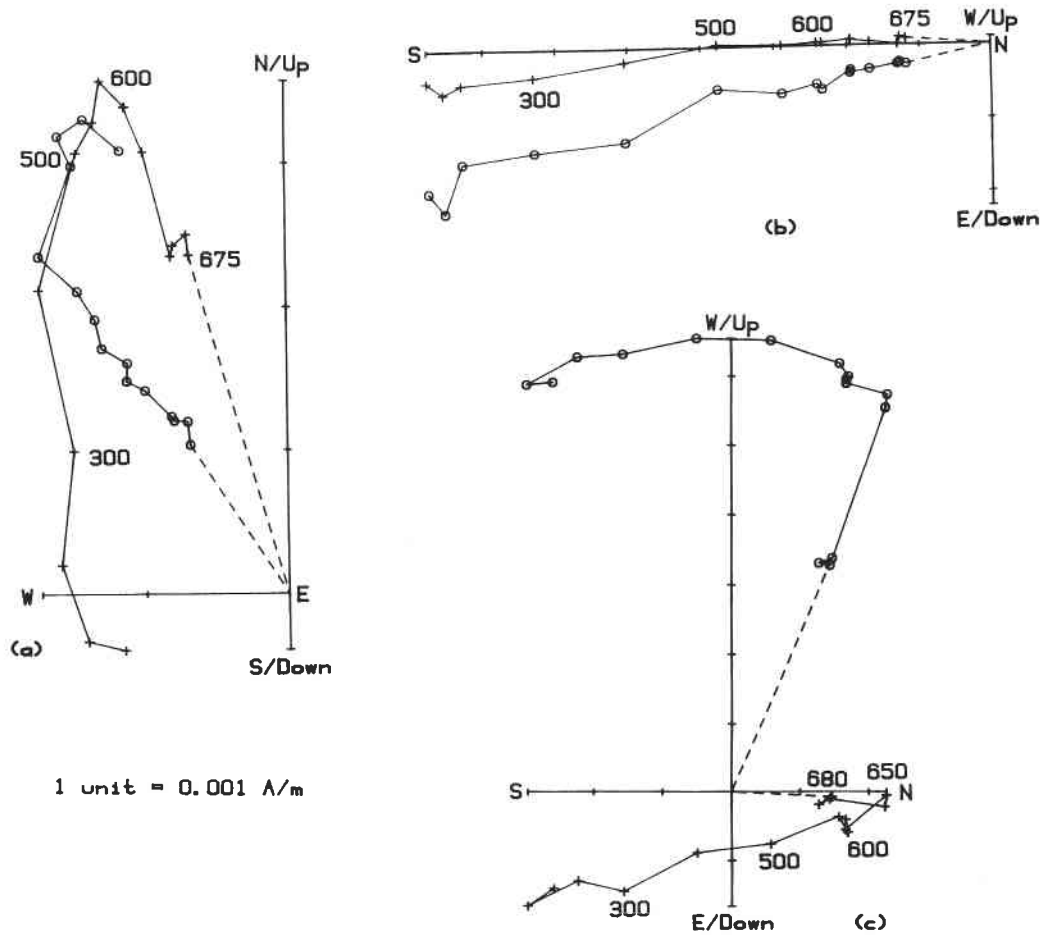


FIG. 2. Maringouin formation demagnetization behavior viewed after tilt correction. (*a*) Sample 60: normal. (*b*) Sample 90: reversed. (*c*) Sample 100: normal. Crosses represent the horizontal component; circles, the vertical component.

(Kirschvink 1980) by computer on linear segments of data chosen by eye from orthogonal plots. High-temperature (C) components that decayed toward the origin were anchored to the origin. We then subjected the low- and high-temperature principal components to Fisher statistics (Fisher 1953) to calculate formation-mean declinations, inclinations, precision parameters (k), and α_{95}.

C component

The choice of principal-component data on which to perform statistical analyses was based on clearer decay trajectories to the origin at high temperature and maximum angular-deviation (MAD) values of less than 15 (only one sample in the Maringouin Formation was chosen with MAD > 10). Of 143 south-limb samples from Maringouin Formation, 103 were accepted; 64 of 92 south-limb samples from the Shepody Formation were also accepted.

We performed a reversal test on the C component by comparing the normal versus reversed directions for the Maringouin and Shepody formations. Mean normal and reversed directions were inverted to a common hemisphere.

We performed fold tests on the high-temperature components of both the Maringouin and Shepody formations by combining the north- and south-limb means and applying Fisher statistics before and after tilt correction.

B component

We used incremental fold tests (Schwartz and Van der Voo 1984; Kent and Opdyke 1985) to find the percentage of unfolding at which k is maximized in the combined (north- and south-limb) data set.

Magnetic stratigraphy

We normalized sample virtual geomagnetic poles (VGP's) by calculating the formation-mean pole and calculating the angular distance of each sample VGP from the mean.

Results

Typically, the measured intensity of the NRM is of the order of 10^{-3}–10^{-2} A/m. Sample NRM declinations are almost entirely reversed. Upon progressive thermal demagnetization most of the samples show some evidence of a northerly-directed magnetization, which is often removed by the first heating at 100°C but may persist in some samples to 300°C. This component (A) is normally seen as a curving trajectory or has too few points to permit principal-component analysis; it was probably acquired in the present geomagnetic field (Figs. 2*a*, 2*c*). Upon the removal of the A component, an intermediate-temperature component (B) is revealed; component B possesses a linear trajectory. This component is sometimes stable to temperatures of 650°C or higher (Fig. 3*a*), though most often the linear segment only persists to 500°–600°C (Figs. 2*b*, 2*c*, 3*c*). The B component is southerly directed and typically has positive inclination. A high-temperature component (C) with both normal and reversed polarity begins to be seen above 550°–600°C in all samples (Figs. 2, 3). This component typically decays toward the origin and is only fully resolved at

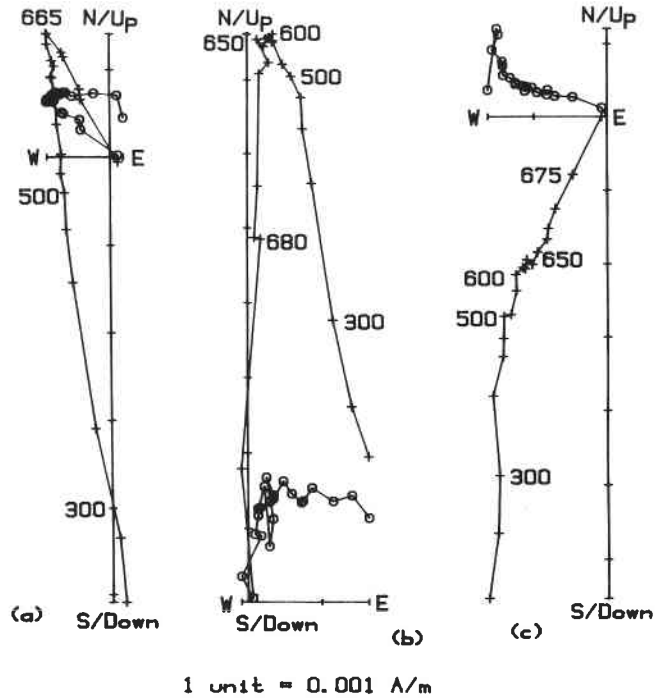


FIG. 3. Shepody Formation demagnetization behavior viewed after tilt correction. (*a*) Sample 225: normal. (*b*) Sample 229: normal. (*c*) Sample 251: reversed. Crosses represent the horizontal component; circles, the vertical component.

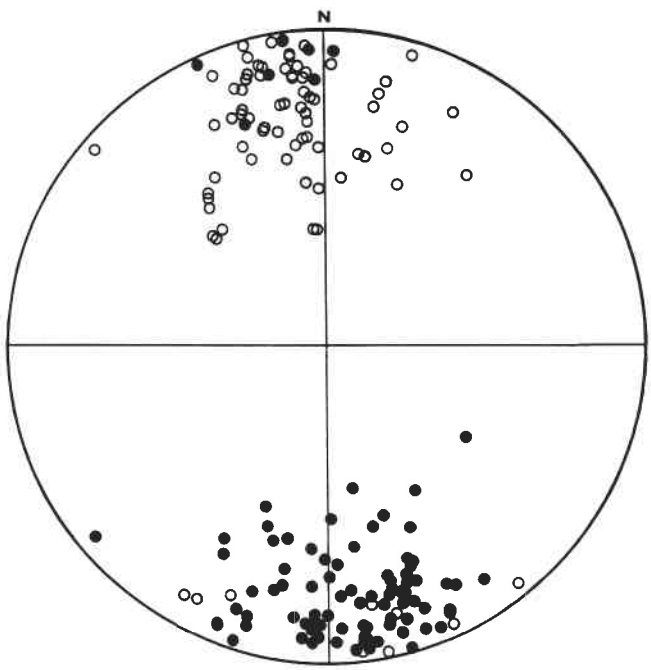


FIG. 4. Equal-area projection of Maringouin and Shepody C-component directions. Solid points represent positive inclination; open points, negative inclination. $N = 167$.

high temperatures. Maximum blocking temperatures are in the 675°–685°C range.

C component

The directions determined by principal-component analysis for the C component are shown for the Maringouin and Shepody formations in Fig. 4. Two antipodal groups, (*i*) north and up and (*ii*) south and down, with some degree of scatter, are seen. All

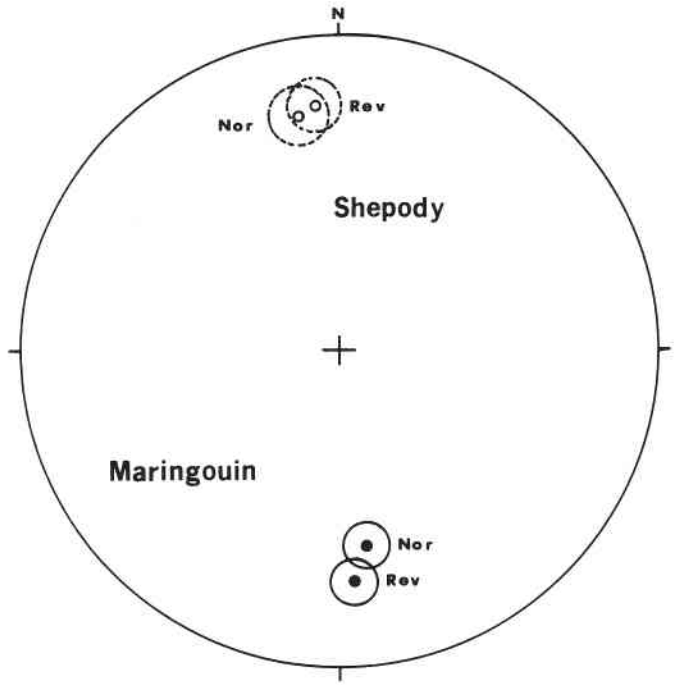


FIG. 5. Reversal tests for the Maringouin and Shepody formations comparing the C-component normal and reversed means inverted to common hemispheres plotted on an equal-angle projection. Solid points represent positive inclination; open points, negative inclination.

TABLE 1. C-component reversal test

Polarity	D (°)	I (°)	α_{95} (°)	k	N	R
		Maringouin Formation (South limb)				
Normal	352.3	−26.2	6.3	11.8	47	43.1
Reversed	176.5	17.9	5.4	13.2	56	51.8
Mean	354.6	−21.7	4.2	12.2	103	94.7
		Shepody Formation (South limb)				
Normal	350.4	−16.3	6.7	16.9	29	27.3
Reversed	174.3	14.4	6.3	15.8	35	32.8
Mean	352.5	−15.2	4.5	16.4	64	60.2

NOTES: D, declination; I, inclination; α_{95}, 95% confidence interval; k, Fisher's precision parameter; N, number of samples; R, length of resultant vector.

points are from the south limb. One hundred and sixty-seven points are plotted in Fig. 4.

In the reversal test, the circles of confidence for the mean normal and reversed directions overlap at the 95% confidence level for both formations (Fig. 5). Based on these results, there is no statistically significant difference between the normal and reversed C-component directions. Therefore, the normal and reversed C-component data may be combined when mean directions are calculated. Table 1 shows the normal- and reversed-polarity directions and the combined means for the south limb. It should be noted that the Maringouin Formation mean is steeper than the Shepody Formation mean by 6.5°, although the circles of confidence overlap. This could be the result of northward plate motion during the time of deposition (northward was probably the drift direction for North America

TABLE 2. C-component fold test

Coord.	D (°)	I (°)	α_{95} (°)	k	N	R
Maringouin Formation (North and south limbs combined)						
Field	2.5	−39.7	7.9	3.5	126	90.7
Tilt	355.0	−22.1	3.8	12.2	126	115.7
Shepody Formation (North and south limbs combined)						
Field	1.7	−41.4	9.7	3.7	80	58.4
Tilt	352.8	−15.6	4.6	13.1	80	74.0

NOTES: Coord., field or tilt coordinates; other abbreviations as in Table 1.

TABLE 3. B-component fold test

Coord.	D (°)	I (°)	α_{95} (°)	k	N	R
Maringouin Formation (South limb)						
Field	170.8	47.6	4.0	48.2	27	26.5
Tilt	168.0	9.5	4.5	38.9	27	26.3
(North limb)						
Field	171.3	−18.7	5.6	32.1	22	21.3
Tilt	171.6	39.3	5.7	30.3	22	21.3
Shepody Formation (South limb)						
Field	173.1	50.0	2.9	35.2	70	35.2
Tilt	167.7	8.0	3.0	32.0	70	67.8
(North limb)						
Field	178.9	−21.7	6.9	14.9	31	29.0
Tilt	181.6	35.5	6.9	14.9	31	29.0

NOTE: Abbreviations as in Tables 1 and 2.

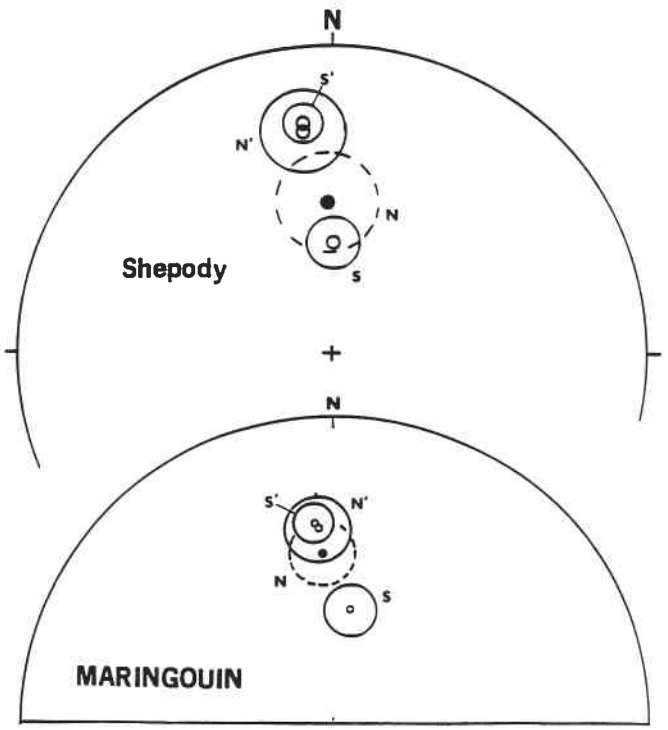


FIG. 6. Fold tests for the Maringouin and Shepody formations. The north- and south-limb mean directions are plotted on equal-angle projections before (N and S) and after (N′ and S′) tilt correction. Solid points represent positive inclination; open points, negative inclination.

in the late Paleozoic). Larson *et al.* (1982) considered the absence of an inclination trend as evidence against the preservation of primary magnetization in the Moenkopi redbed sequence. This subject will be discussed in detail in a paper by DiVenere and Opdyke (in preparation).

Precision (k) increases from 3.5 to 12.2 in the Maringouin Formation and from 3.7 to 13.1 in the Shepody Formation after tilt correction (Table 2). The circles of confidence (α_{95}) similarly improve. The north- and south-limb C-component means for the two formations are plotted before and after tilt correction in Fig. 6. The south-limb directions are steeply up before tilt correction, and the north-limb directions are less steeply down. After tilt correction the north- and south-limb means are nearly indistinguishable. The fold tests for both formations are significant at the 99% confidence level according to the criteria of McElhinny (1964). Although McFadden and Jones (1981) criticized McElhinny's fold test and proposed another test to replace it, McElhinny's test is more stringent and therefore conclusively supports a pre-folding acquisition for the high-temperature component in this study.

B component

The intermediate-temperature B-component north- and south-limb mean directions are dissimilar both before and after tilt correction. The Maringouin and Shepody B-component mean directions are summarized in Table 3. It can be seen that the B component is similar for both formations, unlike the C component, which is significantly different.

The Maringouin Formation B-component directions before (*in situ*) and after tilt correction are plotted on equal-area projections in Fig. 7. The north- and south-limb groups (N and S in Fig. 7) cross during tilt correction: the north-limb inclinations move from shallow up to steeply down, whereas the south-limb inclinations move from steeply down to shallow.

Other workers have previously documented the presence of syn-folding magnetizations in Paleozoic rocks of the folded Appalachians (Kent and Opdyke 1985; Kent 1988). It can be inferred from the crossing inclinations that the B component was acquired during the folding of the beds. The results of the incremental fold tests are shown graphically in Fig. 8. Maximum precision (k) is achieved at 68% of unfolding for the Maringouin Formation and at 71% of unfolding for the Shepody Formation. The resultant mean directions are $D = 169.7°$, $I = 21.3°$, $\alpha_{95} = 3.4°$ for the Maringouin Formation B component and $D = 173.1°$, $I = 19.4°$, $\alpha_{95} = 4.4°$ for the Shepody Formation B component. The north- and south-limb B-component directions from the Maringouin Formation corrected for 68% of tilt are plotted on an equal-area projection in Fig. 7. The clustering of points is greatly improved compared with the *in situ* or tilt-corrected directions.

It should be noted that the mean inclination for the B component when corrected for the percentage of tilt described above yields a steeper inclination than expected. This problem will be discussed in terms of multistage folding or some form of asymmetrical folding in DiVenere and Opdyke (in preparation).

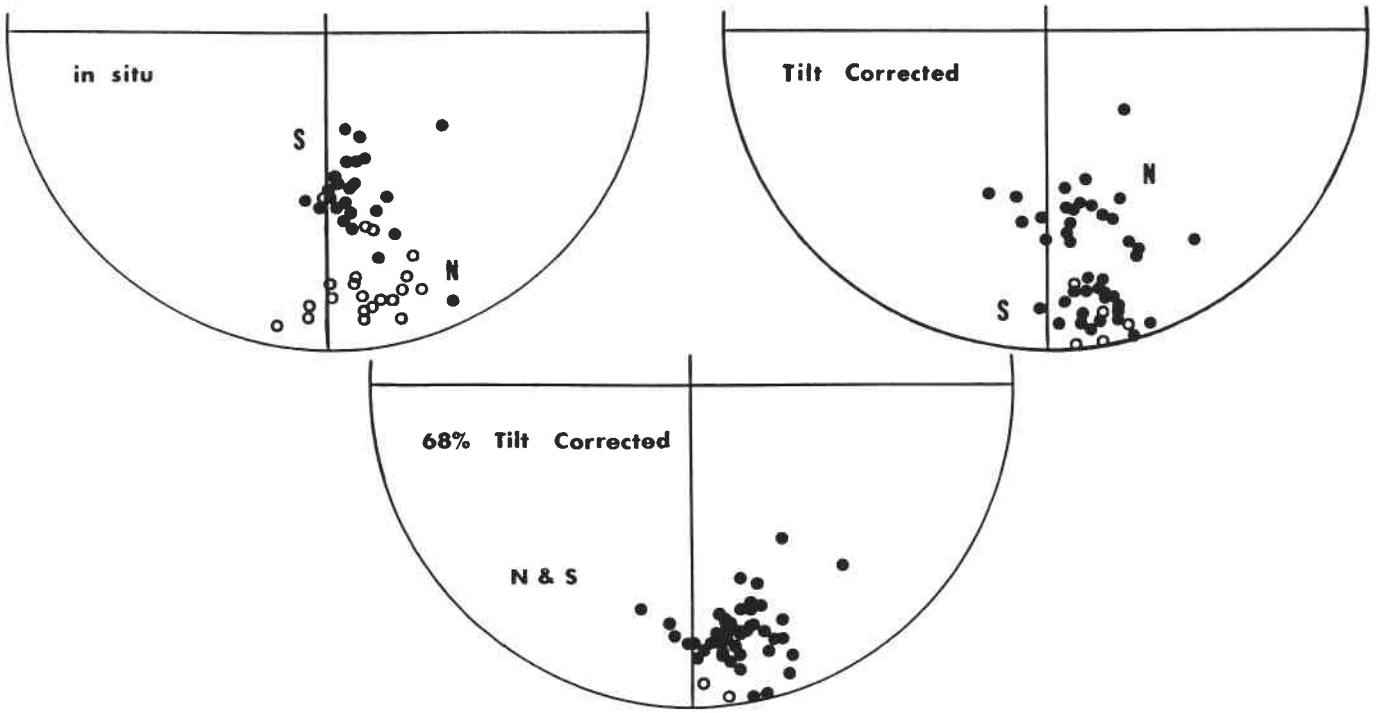


FIG. 7. B component for the Maringouin Formation, with the north (N) and south (S) limbs plotted on equal-area projections before and after tilt correction and after 68% tilt correction.

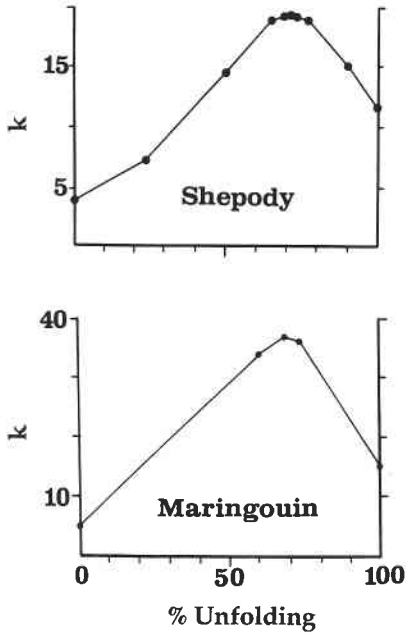


FIG. 8. Incremental fold tests for the Maringouin and Shepody formations, comparing Fisher's precision parameter (k) vs. percentage of unfolding.

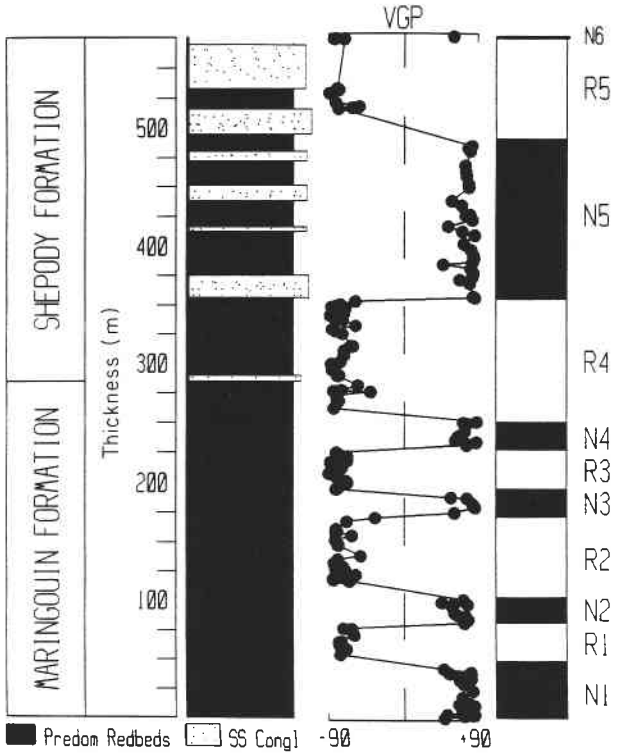


FIG. 9. Magnetostratigraphic column for the Maringouin and Shepody formations. VGP's are normalized relative to formation means.

Magnetic stratigraphy

Plotting normalized sample VGP's versus stratigraphic position yields five discrete reversed-polarity and five discrete normal-polarity zones, with another magnetozone indicated by a single normal sample at the top of the section. The stratigraphic section and magnetic-polarity column are shown in Fig. 9. The polarity is normal at the base of the measured section in the Maringouin Formation, and it is normal at the top, based on one sample, at the Shepody–Enragé contact. The polarity balance is roughly half normal and half reversed. We here refer to normal magnetozones N1–N6 and reversed magnetozones R1–R5 from lowest to highest stratigraphically.

Discussion

The high-temperature component (C) must have been acquired during the deposition or early diagenesis of the sediments. This is clear from its pre-folding acquisition and recorded polarity reversals. It is judged by the red color of the samples and the very high unblocking temperature of the characteristic component that the magnetic carrier is hematite — an early chemical remanent magnetization (CRM) precipitate or detrital remanent magnetization (DRM) hematite. Park (1970) found by X-ray powder methods that hand-picked black heavy-mineral extracts as well as composite grains of red material from two samples from the Maringouin and Shepody formations were composed of α-Fe_2O_3. The remanence was probably acquired during deposition or early diagenesis (Tauxe *et al.* 1980; Tauxe and Badgley 1984). The magnetic stratigraphy and pole positions obtained from the two formations were probably acquired entirely in the Namurian, beginning with Namurian A for the Maringouin Formation and completed before the deposition of the overlying Westphalian A Riversdale Group.

The intermediate-temperature component (B) appears to be a later (syn-folding) CRM, also carried by hematite, judging by the relatively high (600°–650°C) unblocking temperature. The folding, according to Gussow (1953), is Pennsylvanian (Westphalian B) and followed the deposition of the Cumberland Group (Belt 1965).

Roy and Robertson (1968) produced a paleomagnetic pole of 34°N, 117°E for the Maringouin formation; and Roy and Park (1969) produced a paleopole of 34°N, 119°E for the Shepody Formation. Roy and Park (1974) produced a paleopole of 36°N, 122°E for the Maringouin and Shepody formations together. They considered this more accurate than their previous determinations. The present results yield a north paleopole position at 32.4°N, 121.3°E for the Maringouin Formation and 35.7°N, 124.3°E for the Shepody Formation. These results are in close agreement with results by Kent and Opdyke (1985). They obtained a mean paleopole of 32.0°N, 124.8°E from the pre-folding component of the Chesterian Mauch Chunk Formation of Pennsylvania. The Mauch Chunk red sediments were deposited in the latest Mississippian, probably overlapping the time of deposition of the Maringouin Formation. The Mauch Chunk results are complicated by oroclinal rotation about the Pennsylvania salient as documented by Kent (1988). Irving and Strong (1984), from the Visean Deer Lake Group red sediments of Newfoundland, obtained a pre-fold paleopole at the same longitude as the above studies but 10° farther south.

Figure 10 shows a portion of the Paleozoic polar wandering path with the paleopoles for the Maringouin, Shepody, Mauch Chunk, and Maroon–Minturn formations and the Morien and Deer Lake groups, as well as 30 Ma sliding window mean poles from Irving and Irving (1982). The directional change between the Early and Middle Carboniferous results and the late Paleozoic mean poles of Irving and Irving (1982) may indicate a major collisional event in the closing of the proto-Atlantic shortly before 300 Ma ago.

Conclusions

This study has shown that the pre-tectonic component of magnetization can be resolved in Carboniferous red sediments by progressive thermal demagnetization to temperatures of 685°C or greater and that a coherent and detailed magnetic stratigraphy can be constructed. The mean direction of the characteristic component of magnetization is $D = 355.0°$, $I = -22.1°$, $\alpha_{95} = 3.8°$ for the Maringouing Formation and $D = 352.8°$, $I = -15.6°$, $\alpha_{95} = 4.6°$ for the Shepody Formation. These results yield north paleopole positions at 32.4°N, 121.3°E for the Maringouin Formation and 35.7°N, 124.3°E for the Shepody Formation. The 6.5° shallowing in inclination between the Maringouin and Shepody mean directions could be documenting northward plate motion. Plotting sample declinations versus stratigraphic position yields a Namurian magnetic stratigraphy with five normal-polarity and five reversed-polarity zones and the suggestion of a sixth normal zone at the top of the measured sequence (based on one normal sample).

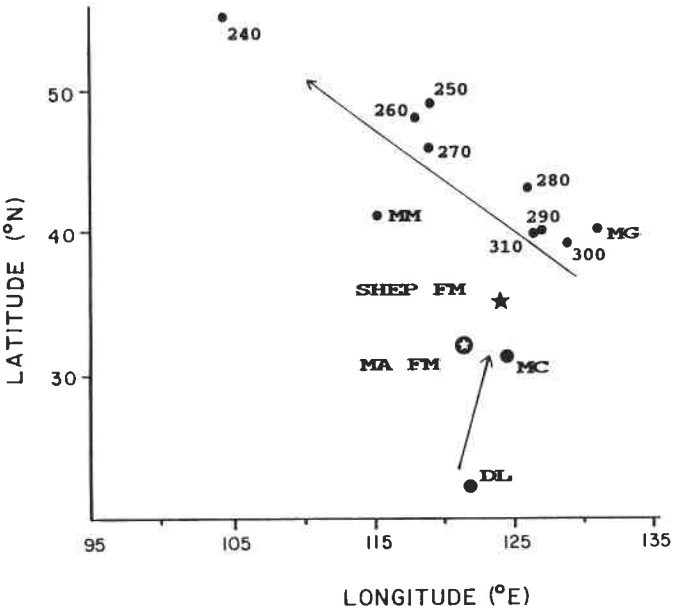


FIG. 10. Late Paleozoic paleopoles: Maringouin Formation (MA FM); Shepody Formation (SHEP FM); Mauch Chunk Formation (MC) (Kent and Opdyke 1985); Deer Lake Group (DL) (Irving and Strong 1984); Morien Group (MG) (Scotese *et al.* 1984); Maroon and Minturn formations (MM) (Miller and Opdyke 1985); and 30 Ma sliding window mean poles (Irving and Irving 1982).

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